Required Capstone Checkpoint 2.1: Retrieval Strategy Design and Baseline Implementation

Directions

You will continue working with the same dataset and scenario you selected for your capstone checkpoint in Module 1. In this activity, you will design and implement a baseline retrieval system that supports the types of queries required for your scenario.

Download this worksheet and save a copy to your device before you begin working. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Use this worksheet to:

- Document your retrieval approach
- Capture representative retrieval queries and results
- Provide evidence that your retrieval system is functioning
- Observe how retrieval performance changes across different query types
- Reflect on where the retrieval approach performs well and where it struggles

1. Selected Scenario and Retrieval Approach

- Identify your chosen capstone scenario.
- Describe the retrieval approach you implemented (keyword-based, semantic, or hybrid).
- Briefly explain why this retrieval approach is appropriate for your dataset and query types.
- Describe any trade-offs you considered when selecting this approach

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Scenario - Wikipedia Retrieval Engine

Retrieval approach used, brief explanation, and description of trade-offs considered
Scenario and objective. I built a retrieval-augmented Wikipedia question-answering system over a local corpus of 2,417 HTML articles. Each question is independent; no conversation history is used. **Retrieval approaches.** I compared BM25 keyword retrieval, Chroma vector search, and weighted hybrid retrieval. The initial baseline used 1,600-character chunks with 200-character overlap and Top K=4. ● BM25 ● Vector Search ● Hybrid with BM25/vector weights of 0.50/0.50, 0.70/0.30, and 0.30/0.70. **Extended-context experiments.** Because many runs found the correct article but did not return enough text, I added two context strategies: HTML-aware section chunks (up to 10,000 characters, no overlap) and full-article expansion using separately stored plain-text articles. I also tested section Top K=10 and full-article Top K=7. The extended hybrid used BM25/vector weights of 0.35/0.65 favoring VS findings, because BM25 more often introduced lexically related but irrelevant articles. **Why this design fits the task.** The questions range from precise fact lookup to broad summaries, time-bounded lists, aggregation, and comparisons across articles. BM25 rewards exact terms and names; vector search handles paraphrases and conceptual similarity; hybrid retrieval can preserve both signals. **Trade-offs considered:** ● BM25 is fast and interpretable but can over-rank lexical matches that are out of context. ● Vector search usually produces more relevant articles, but it requires an embedding model and persistent Chroma index. ● More context improves recall, but full articles and high Top K can add enough irrelevant text to reduce answer synthesis quality. ● Sections approach are more targeted, but a child section can be too narrow unless its parent and sibling sections are also available.

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2. Baseline Retrieval System Evidence

- Provide evidence that your retrieval system was implemented and tested.
- Evidence may include screenshots, logs, outputs, notebook examples, code snippets, or brief descriptions showing that retrieval queries were executed successfully.

Evidence of retrieval system implementation and testing
The system was implemented and exercised across 13 representative questions and multiple retrieval/context configurations. • A shared RetrieverSpec records retriever kind, Top K, hybrid candidate pool, and weights so configurations can be reproduced and compared. • BM25, vector, and hybrid behavior are implemented in separate retriever classes; LLM prompting/calling is isolated in llm.py. • Chroma indexes are persisted under retrievers/db/<chroma_db_name> and reused after restart. Full article text is stored separately under retrievers/db/article_text/wikipedia_plain_text. • Each timestamped evidence log begins with the full configuration and records the query, ranked articles, retrieved chunks/sections and scores, context sent to the LLM, and grounded answer. Log sample is attached. • The attached analysis workbook compares outcomes against a manually compiled ground-truth workbook and separates article retrieval, context sufficiency, and answer synthesis observations. *These artifacts show successful corpus preprocessing, database reuse, retrieval execution, context assembly, LLM invocation, and repeatable evidence collection.*

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3. Representative Queries and Retrieved Results

- Provide three to five representative queries relevant to your scenario.
- Include the retrieved results for each query.
- Focus on showing how the retrieval system behaves across different types of information needs. Where possible, include a mix of precise, conceptual, or broader exploratory queries.

Query 1
Query: According to the Wikipedia article on the 100 meters, what is the current men's world record, who holds it, and when was it set? Quote the article. **Retrieved results:** Every RAG configuration returned the correct answer and quote: 9.58 seconds, Usain Bolt, set on 16 August 2009. Vector and hybrid runs concentrated on the 100 metres article; BM25 also returned unrelated lexical matches such as tall buildings and older Olympics material. **Observations:** This precise, single-article fact lookup was the easiest query type. The answer was stated directly in a relevant passage, so all methods succeeded once the correct article appeared. Vector search showed better precision than BM25, although the LLM ignored BM25's extra noise.

Query 2
Query: List three significant events recorded in the Wikipedia article for the year 1580, quoting the article for each. **Retrieved results:** The 1,600-character baseline usually retrieved the correct 1580 article but exposed only isolated parts of the year, producing weak synthesis. Section retrieval improved coverage when January-March, April-June, and July-September were retrieved. Full-article expansion produced a correct answer with quotations; section Top K=10 also reached a correct result. **Observations:** The ranking problem was mostly solved, but the context was initially incomplete. This query demonstrated that finding an article and supplying enough of that article are different tasks. Natural sections improved focus; full-article expansion improved completeness. LLM did a good job ignoring irrelevant articles

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Query 3
Query: What were the most devastating earthquakes in the last 50 years? Show human deaths for each earthquake. **Retrieved results:** Chunked vector retrieval was only partially correct; BM25 found the top two, while several hybrid runs found the top event but included lower-ranked or out-of-window events. Vector section retrieval produced a correct detailed answer, and all full-article configurations produced comprehensive results. **Observations:** Broad, time-bounded list questions require both enough source coverage and correct filtering. Larger natural sections or full articles supplied the needed tables and lists, but some runs failed to enforce the 50-year boundary. This is partly retrieval/context coverage and partly LLM reasoning over dates.

Optional Additional Queries
Query: Compare countries by their life expectancy versus population. Is there a correlation? **Retrieved results:** No tested configuration retrieved both required datasets. Runs commonly found the life-expectancy article but missed the UN-based population article; full-article runs instead added region-specific articles. Without both tables, the model could not perform the comparison. The ground truth required matching 182 countries and found essentially no correlation. **Observations:** This was the clearest retrieval failure. A single semantic query did not reliably express two separate evidence needs. The question should be decomposed into population retrieval and life-expectancy retrieval, followed by country-name normalization and a deterministic comparison step.

4. Retrieval Performance Reflection

- Reflect on where your retrieval approach performs well and where it struggles.
- Consider how different query types affect retrieval quality, relevance, flexibility, or precision.

Retrieval strengths
The system was strongest on precise, single-article questions whose answer appeared in one passage. • The 100 metres world-record question and both 1972 Winter Olympics questions were answered correctly across nearly every retrieval configuration. • Vector search generally returned more contextually relevant articles than BM25, especially when query wording did not exactly match the source.

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- Full-article expansion was effective for broad single-article synthesis, while section retrieval offered a useful balance between precision and coverage.
- Persistent Chroma indexes and configuration-first logs made repeated experiments comparable without rebuilding the vector database.

Retrieval limitations or challenges

The hardest questions required multiple sources, complete tables, aggregation, or interpretation across distant parts of an article.

- The population/life-expectancy comparison failed because retrieval consistently missed one required source.
- The Academy Award winner query sometimes retrieved sufficient evidence but still counted or synthesized it incorrectly, including an answer that contradicted its own evidence. Retrieval success therefore does not guarantee answer correctness.
- Character chunks fragmented broad answers; section chunks could be too narrow; full articles could overwhelm the LLM when Top K was increased.
- HTML-to-text conversion flattened tables, lists, and legend symbols. This may remove or obscure legend needed to distinguish Academy Award winners, nominees, and special markers.
- Higher Top K increased recall for section retrieval but degraded full-article synthesis by sending more irrelevant text. Top K must be tuned with the context unit, not as a universal setting.

Patterns observed across different query types

Three distinct evaluation layers emerged from the experiments:

- Article recall: Did retrieval find every source required by the question?
- Context sufficiency: Did the selected chunks or sections contain enough of each source?
- Answer synthesis: Did the LLM correctly filter, aggregate, compare, and quote the supplied evidence?

Precise fact questions usually passed all three layers. Broad single-article questions often passed article recall but failed context sufficiency. Comparative and counting questions could fail at article recall or at synthesis even when the evidence was present. Hybrid weights made less difference than context construction because BM25 and vector search

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often found the same main article; the larger gains came from sections, full-article expansion, and Top K appropriate to the context size.

5. Retrieval Strategy Reflection

If you were continuing to improve this system, would you keep the same retrieval approach or make changes? Explain your reasoning.

Reflection
I would keep semantic retrieval as the primary signal and retain a lower-weight BM25 component, but I would change how retrieved evidence is expanded and evaluated. My next baseline would be the 0.35 BM25 / 0.65 vector hybrid over HTML-aware sections, followed by parent/child expansion and article-level deduplication. **Planned improvements:** ● Preserve document structure during HTML preprocessing, especially headings, tables, list boundaries, and legend symbols. ● Retrieve a small candidate set of sections, rerank them, then expand selected children to their parent section or whole article only when broader context is needed. ● Deduplicate by article and cap the number of full articles sent to the LLM so increasing Top K does not multiply context noise. ● Decompose multi-source questions into explicit subqueries, then combine the retrieved datasets with deterministic code before asking the LLM to explain the result. ● Score experiments separately for article recall, context sufficiency, and final-answer correctness instead of treating the final answer as the only metric. *The experiments show that more text is not automatically better. The most promising direction is targeted, structurally complete context: enough surrounding material to answer the question, but not so much unrelated content that the model loses the evidence.*

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Supporting Materials

The submission is supported by the following evidence artifacts:

- Checkpoint 2.1 Retrieval Analysis_Expanded.xlsx - experiment-by-question observations across BM25, vector, hybrid, character chunks, sections, full articles, and higher Top K runs.
- wikipedia Q-A GT.xlsx - manually compiled expected answers used for qualitative comparison.
- Timestamped evidence logs - configuration snapshots, ranked article/chunk evidence, context sent to the LLM, and grounded answers for each independent question.

AI Assistance Disclosure

I used ChatGPT to assist with organizing my code and developing the testing framework, including parsing HTML, generating comprehensive logs, and creating a framework for running multiple experiments.

I also used ChatGPT to organize my observations and refine the wording of my written analysis. I reviewed the prompts, experimental runs, interpretations, and final conclusions and made all final decisions regarding their inclusion.

End of Required Capstone Checkpoint 2.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.

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